



ECE 3340 Quiz #1

Name: Corinna LevyMYUH#: 2301734

1. (10 pts) Demonstrate the following 8-bit binary unsigned integer operation. Provide the corresponding operands and the results in decimal. Please show your work.

$$1010\ 0101_2 \times 10_2 = 165_{10} \times 2_{10}$$

$$\begin{array}{r} \text{binary} \\ 1010\ 0101 \\ \times \quad 10 \\ \hline 1\ 0100\ 1010 \end{array}$$

$$\boxed{1\ 0100\ 1010_2}$$

$$\text{decimal } 1010\ 0101_2 = 2^6 + 2^5 + 2^4 + 2^3 = 1+4+32+128=165$$

$$\begin{array}{r} 165 \\ \times 2 \\ \hline 330 \end{array} \quad \boxed{330_{10}}$$

$$1\ 0100\ 1010_2 = 2^7 + 2^6 + 2^5 + 2^4 = 2+8+64+256=330_{10} \quad \checkmark$$

2. (10 pts) Convert the following 8-bit signed integers into decimal using Two's complement or convert the following decimal into 8-bit signed integers using Two's complement. Please show your work.

$$\begin{array}{r} \text{negative} \\ 1011\ 1000 \\ + 0100\ 0111 \\ \hline 0100\ 1000_2 \end{array}$$

$$0100\ 1000_2 = 2^3 + 2^4 = 72_{10}$$

$$\boxed{-72_{10}}$$

$$-85_{10} \quad 85_{10} = 55_{16} = 0101\ 0101_2$$

$$\begin{array}{r} 16 \overline{) 85} \\ \underline{-80} \\ 5 \end{array}$$

$$\begin{array}{r} 1010\ 1010 \\ + \quad 1 \\ \hline 1010\ 1011_2 \end{array}$$

3. (8 pts) Convert the numbers into the specified number system.

1

- 1) Convert $(1023)_{10}$ to base-8 (Radix = 8);

$$1023_{10} = (3 \times 8^2) + (2 \times 8^1) + (3 \times 8^0) = 3 \times 10 + 125 = 138_{10}$$

$$\begin{array}{r} 17 \\ 8 \overline{) 1023} \\ \underline{-80} \\ 223 \\ \underline{-160} \\ 63 \\ \underline{-56} \\ 7 \end{array}$$

$$1023_{10} = \boxed{212_8}$$

- 2) Convert $(762)_{10}$ to base-5;

$$762_{10} = (2 \times 8^2) + (6 \times 8^1) + (7 \times 8^0) = 2 \times 48 + 448 = 448_{10}$$

$$\begin{array}{r} 14 \\ 5 \overline{) 762} \\ \underline{-50} \\ 262 \\ \underline{-250} \\ 12 \\ \underline{-10} \\ 2 \end{array}$$

$$762_{10} = \boxed{3443_5}$$

4. (10 pts) Convert the following values (unsigned) to decimal. Please show your work.

$$0111.1011 = 2^2 + 2^1 + 2^0 + 2^{-1} + 2^{-3} + 2^{-4} = 4 + 2 + 1 + \frac{1}{2} + \frac{1}{8} + \frac{1}{16} = \boxed{7.6875_{10}}$$

$$10101.101 = 2^4 + 2^2 + 2^0 + 2^{-1} + 2^{-3} = 16 + 4 + 1 + \frac{1}{2} + \frac{1}{8} = \boxed{21.625_{10}}$$

5. (10 pts) Assume an 8-bit floating point format with a 4-bit exponent and a bias of 7 and implied 1. Provide the decimal representations of the following stored values.

fraction led by 1
subtract by 7 for exponent

Sign	Exponent	Mantissa
1	1 1 0 0	1 0 1

negative

sign, 2^{exponent}, mantissa

$$-1 \cdot 2^{1100_2} \cdot 1.101_2$$

Bias 7:

$$12 - 7 = 5 \rightarrow 2^5 = 32$$

$$-1 \cdot 32 \cdot 1.625 = \boxed{-52_{10}}$$

$$1.101_2 = 1 + \frac{1}{2} + \frac{1}{8} = 1.625_{10}$$

2

Sign	Exponent	Mantissa
0	0 1 0 1	0 1 0

positive

$$1 \cdot 2^{0101_2} \cdot 1.010_2$$

Bias 7:

$$5 - 7 = -2 \rightarrow 2^{-2} = \frac{1}{4}$$

$$1.010_2 = 1 + \frac{1}{4} = 1.25_{10}$$

$$1 \cdot \frac{1}{4} \cdot 1.25 = \boxed{0.3125_{10}}$$

6. (5 pts) What is the computational complexity of multiplying two $n \times n$ matrices using a naive algorithm?

The computational complexity of multiplying 2 matrices is $O(n^3)$.

7. (12 pts) Reformulate the following expression to deal with catastrophic cancellation and test the result using 3 digits of precision for $x = 0.01$.

$$\sqrt[4]{1+x} - \cos(x)$$

at small x , $\sqrt[4]{1+x} \approx \cos(x)$, so w/ 3 sig figs, they cancel

Exact Result (using calculator or computer)

$$\text{At } x=0.01: \sqrt[4]{1+0.01} - \cos(0.01) = \boxed{0.0025406788766}$$

Reformulation

$$a-b = \frac{(a-b)(a+b)}{(a+b)} = \frac{(a^2-b^2)(a^2+b^2)}{(a+b)(a^2+b^2)} = \frac{a^4-b^4}{(a+b)(a^2+b^2)} = \frac{1+x-\cos^4(x)}{(\sqrt[4]{1+x}+\cos(x))(\sqrt[4]{1+x}+\cos^2(x))}$$

Original result

$$\sqrt[4]{1+0.01} - \cos(0.01)$$

$$= 1.00 - 1.00$$

$$= \boxed{0.00}$$

Reformulated result

$$\frac{1+0.01-\cos^4(0.01)}{(\sqrt[4]{1+0.01}+\cos(0.01))(\sqrt[4]{1+0.01}+\cos^2(0.01))}$$

$$= \frac{1.01-1.00}{(1.00+1.00)(1.00+1.00)} = \frac{0.01}{4.00}$$

$$= \boxed{0.0025}$$

3

Relative error

$$\% \text{ error} = \frac{|1 - 0.0025406788766|}{0.0025406788766} \times 100\%$$

$$= \boxed{100\%}$$

Relative error

$$\% \text{ error} = \frac{|0.0025 - 0.0025406788766|}{0.0025406788766} \times 100\%$$

$$= \boxed{1.60\%}$$

8. (10 pts) Reformulate using the Mean Value Theorem and test the result using 3 digits of precision for $x = 0.01$. Show your work.

$$\sqrt{\pi+x} - \sqrt{\pi}$$

Exact Result (using calculator or computer)

$$\text{At } x=0.01: \sqrt{\pi+0.01} - \sqrt{\pi} = \boxed{0.002818706644}$$

Reformulation

$$a-b = \frac{(a-b)(a+b)}{(a+b)} = \frac{a^2-b^2}{a+b} = \frac{\pi+x-\pi}{\sqrt{\pi+x}+\sqrt{\pi}} = \frac{x}{\sqrt{\pi+x}+\sqrt{\pi}}$$

Original result

$$\sqrt{\pi+0.01} - \sqrt{\pi}$$

$$= 1.78 - 1.77$$

$$= \boxed{0.01}$$

Reformulated result

$$\frac{0.01}{\sqrt{\pi+0.01}+\sqrt{\pi}}$$

$$= \frac{0.01}{1.78+1.77}$$

$$= \boxed{0.002818706644}$$

Relative error

$$\% \text{ error} = \frac{|0.01 - 0.002818706644|}{0.002818706644} \times 100\%$$

$$= \boxed{255\%}$$

Relative error

$$\% \text{ error} = \frac{|0.002818706644 - 0.002818706644|}{0.002818706644} \times 100\%$$

$$= \boxed{0.064\%}$$

9. (10 pts) Use a Taylor series expansion to reformulate the following numerical problem. Keep terms up to and including $O(x^3)$. Calculate the relative error

4

before and after reformulation for $x = 0.02$. Use 3 significant digits for each operation.

$$\frac{1 - \cos x}{x}$$

Exact Result (using calculator or computer)

$$\text{At } x=0.02: \frac{1 - \cos(0.02)}{0.02} = \boxed{0.00999966667111}$$

Reformulation

$$\text{For } \cos(x): T(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \quad \text{For } 1 - \cos(x): T(x) = \frac{x^2}{2!} - \frac{x^4}{4!} + \frac{x^6}{6!} + \dots$$

$$\text{For } \frac{1 - \cos(x)}{x}: T(x) = \frac{x^2}{2!} - \frac{x^3}{4!} + \frac{x^5}{6!} + \dots$$

$$\text{Keeping up to 3 including } O(x^3): T(x) = \frac{x^2}{2} - \frac{x^3}{24}$$

Original result

$$\frac{1 - \cos(0.02)}{0.02}$$

$$= \frac{1.00 - 1.00}{0.02}$$

$$= \boxed{0.00}$$

Reformulated result

$$\frac{0.02}{2} - \frac{(0.02)^3}{24}$$

$$= \frac{0.02}{2} - \frac{0.009}{24}$$

$$= \boxed{0.01}$$

Relative error

$$\% \text{ error} = \frac{|0.01 - 0.00999966667111|}{0.00999966667111} \times 100\%$$

$$= \boxed{100\%}$$

Relative error

$$\% \text{ error} = \frac{|0.01 - 0.00999966667111|}{0.00999966667111} \times 100\%$$

$$= \boxed{0.0033\%}$$

10. (15 pts) Use synthetic division to compute $f(x) = e^{-x} \sin x^2$ for $x = 0.1$ with an error bound of $O(x^4)$ using 3 significant digits. Show this calculation using synthetic division.

$$\text{For } e^{-x}: T(x) = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots \quad \text{For } e^{-x}: T(x) = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots$$

$$\text{For } \sin(x): T(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \quad \text{For } \sin(x): T(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\text{For } e^{-x} \sin(x): T(x) = (1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \dots)(x^2 - \dots) = x^2 - x^3 + \frac{x^4}{2} - \frac{x^6}{6} + \dots$$

$$T(x) = x^2 - x^3 + O(x^4)$$

5

$$\text{Horner: } \begin{array}{r} -1 \quad 1 \quad 0 \quad 0 \\ \quad -x \quad x(1-x) \quad x(x(1-x)) \\ -1 \quad 1-x \quad x(1-x) \quad x(x(1-x)) \end{array}$$

$$\text{For } x=0.1: ((1-0.1)0.1)0.1$$

$$((0.9)0.1)0.1$$

$$(0.09)0.1$$

$$\boxed{0.009}$$

$$\text{Check: } f(0.1) = 0.009048223375$$

$$\% \text{ error} = \frac{|0.009 - 0.009048223375|}{0.009048223375} \times 100\% = 0.53\% \quad \checkmark$$

6